What each head learned, and whether the gate noticed

Rows sorted by the head that wins them · specialisation is emergent: no head was told which domain or style to own

Head skill by source

NLL gap vs average head (nats, + = better)

sources	Head 0	Head 1	Head 2
LexGLUE / ledgar	**+0.394**	-0.174	-0.220
LegalCaseDocumentSummarization / …	**+0.039**	-0.027	-0.012
MiningLegalArguments / argType	**+0.023**	-0.007	-0.016
MiningLegalArguments / agent	**+0.021**	-0.018	-0.003
CaseBriefs / case_briefs	**+0.021**	-0.025	+0.005
NaturalInstructionsLegal / billsu…	**+0.014**	-0.019	+0.005
StackExchangeQuestionsLegal / sta…	**+0.012**	-0.003	-0.009
EdgarNER /	**+0.009**	-0.008	-0.001
MultiLexSum / long_to_tiny	-0.031	**+0.033**	-0.002
MAUD / text_type	-0.003	**+0.015**	-0.012
NaturalInstructionsLegal / caseho…	+0.003	**+0.009**	-0.013
EurLexSum / english	+0.014	-0.053	**+0.038**
RedditLegalQA / reddit_legal_qa	-0.001	-0.006	**+0.007**
IndianNER /	+0.005	-0.011	**+0.006**

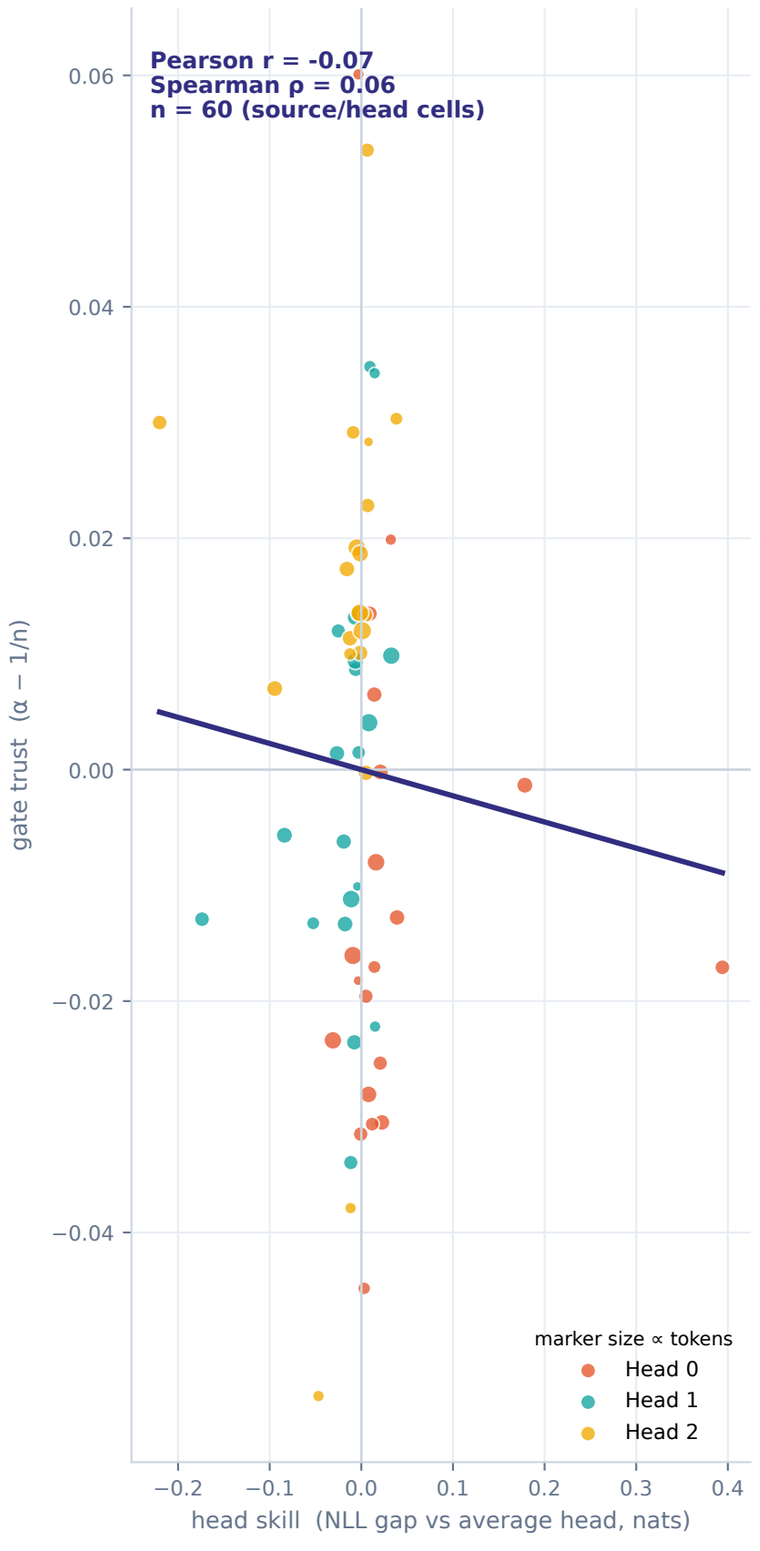
Gate trust by source

mean α given to that head

Head 0	Head 1	Head 2
32%	32%	**36%**
32%	33%	**34%**
30%	35%	**35%**
33%	32%	**35%**
31%	35%	**35%**
34%	33%	33%
30%	33%	**36%**
35%	31%	34%
31%	34%	**35%**
39%	31%	30%
29%	**37%**	34%
32%	32%	**36%**
30%	34%	**36%**
31%	30%	**39%**

Does the gate follow real skill?

up-and-right = gate trusts heads that are actually better



Head skill by task type

task types	Head 0	Head 1	Head 2
TEXT_CLASSIFICATION	**+0.178**	-0.084	-0.095
NATURAL_LANGUAGE_INFERENCE	**+0.032**	+0.015	-0.047
NAMED_ENTITY_RECOGNITION	**+0.016**	-0.011	-0.005
QUESTION_ANSWERING	**+0.008**	-0.007	-0.001
SUMMARIZATION	-0.009	**+0.008**	+0.001
MULTIPLE_CHOICE	-0.003	-0.004	**+0.008**

Gate trust by task type

Head 0	Head 1	Head 2
33%	33%	**34%**
35%	**37%**	28%
33%	32%	**35%**
31%	34%	**35%**
32%	34%	**35%**
32%	32%	**36%**